xgboost

- Liad Mandil, 315096073
- Maxim Prokopchuk, 337875397

Imports

In [61]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.cm as cm
from sklearn.model_selection import train_test_split , cross_val_score, Stratifi
from xgboost import XGBClassifier
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score, classification_report, ConfusionMatr
from sklearn import tree
from sklearn.model_selection import train_test_split , cross_val_score, Stratifi
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_sc
from sklearn.metrics import roc_curve, auc
```

Load Data

In [62]:
```
spaceship_data = pd.read_csv('final_data.csv')
```

split to x and y

In [63]:
```
y = spaceship_data["Transported"].astype(int)
x = spaceship_data.drop(columns=["Transported"])
```

Train-Test Split

In [64]:
```
x_train, x_test, y_train, y_test = train_test_split(
    x, y, test_size=0.2, random_state=42
)
```

In [65]:
```
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
```

בדיקת ערכי ההייפר פרמטרים

Base

In [66]:
```
xgb_base = XGBClassifier(
    random_state=42,
    n_estimators=300,
```

```
    max_depth=4,
    learning_rate=0.1,
)

fold_scores = cross_val_score(
    xgb_base,
    x_train,
    y_train,
    cv=cv,
    scoring="accuracy",
    n_jobs=-1
)

print("XGBoost Baseline")
print("Fold accuracies (%):", np.round(100 * fold_scores, 2))
print(f"Mean CV accuracy: {100*fold_scores.mean():.2f}% | Std: {100*fold_scores.
```

```
XGBoost Baseline
Fold accuracies (%): [79.78 80.07 81.01 80.79 80.2 ]
Mean CV accuracy: 80.37% | Std: 0.46%
```

max_depth

In [67]:

```
results_depth = []
best_depth = None
best_depth_score = -1

depth_list = [2, 3, 4, 5, 6, 7]

for depth in depth_list:

    xgb = XGBClassifier(
        random_state=42,
        n_estimators=300,
        max_depth=depth,
        learning_rate=0.1,
        subsample=1.0,
        colsample_bytree=1.0,
        eval_metric="logloss",
        n_jobs=-1
    )

    fold_scores = cross_val_score(
        xgb,
        x_train,
        y_train,
        cv=cv,
        scoring="accuracy",
        n_jobs=-1
    )

    mean_score = fold_scores.mean()

    results_depth.append({
        "max_depth": depth,
        "cv_mean": mean_score,
        "cv_std": fold_scores.std()
    })

    if mean_score > best_depth_score:
```

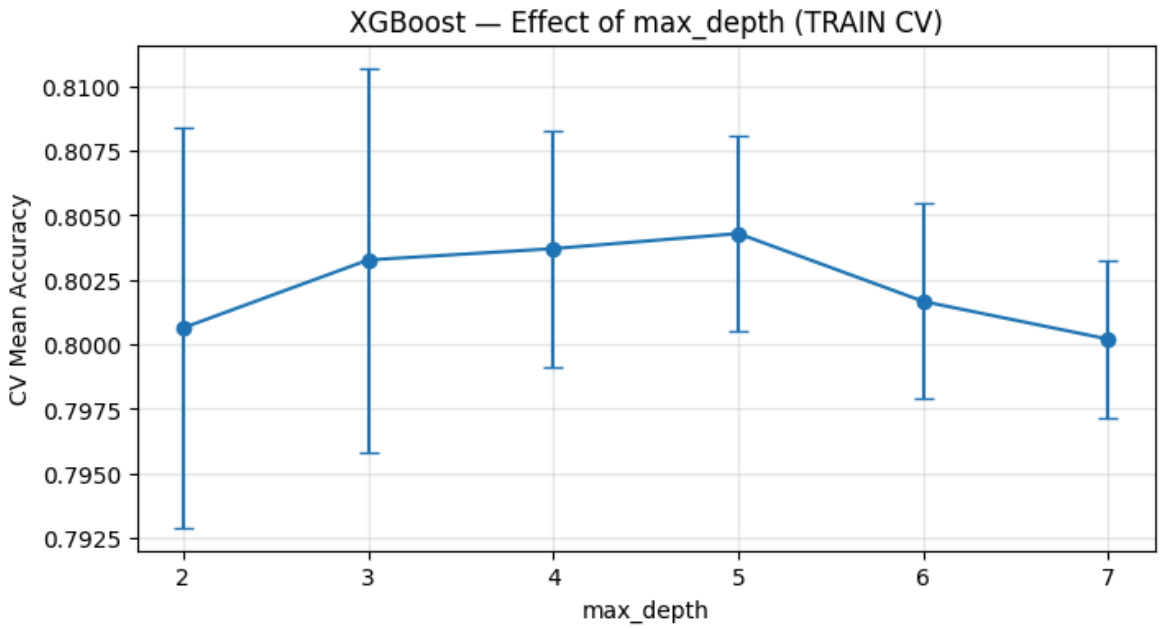
```
        best_depth_score = mean_score
        best_depth = depth
```

In [68]:
```
results_depth_df = pd.DataFrame(results_depth).sort_values("cv_mean", ascending=
display(results_depth_df.round(4))

print("Best depth:", best_depth)
```

	max_depth	cv_mean	cv_std
3	5	0.8043	0.0038
2	4	0.8037	0.0046
1	3	0.8033	0.0074
4	6	0.8017	0.0038
0	2	0.8006	0.0077
5	7	0.8002	0.0030

```
Best depth: 5
```

In [69]:
```
tmp = results_depth_df.sort_values("max_depth")

plt.figure(figsize=(8,4))
plt.errorbar(
    tmp["max_depth"],
    tmp["cv_mean"],
    yerr=tmp["cv_std"],
    fmt="o-",
    capsize=4
)

plt.title("XGBoost — Effect of max_depth (TRAIN CV)")
plt.xlabel("max_depth")
plt.ylabel("CV Mean Accuracy")
plt.grid(alpha=0.3)
plt.show()
```



n_estimators

In [70]:
```
results_trees = []
best_trees = None
best_trees_score = -1

n_list = range(100, 1001, 50)

for n in n_list:

    xgb = XGBClassifier(
        random_state=42,
        n_estimators=n,
        max_depth=best_depth,
        learning_rate=0.1,

    )

    fold_scores = cross_val_score(
        xgb,
        x_train,
        y_train,
        cv=cv,
        scoring="accuracy",
        n_jobs=-1
    )

    mean_score = fold_scores.mean()

    results_trees.append({
        "n_estimators": n,
        "cv_mean": mean_score,
        "cv_std": fold_scores.std()
    })

    if mean_score > best_trees_score:
        best_trees_score = mean_score
        best_trees = n
```

In [71]:
```
results_trees_df = pd.DataFrame(results_trees).sort_values("cv_mean", ascending=
display(results_trees_df.round(4).head(5))

print("Best n_estimators:", best_trees)
```

	n_estimators	cv_mean	cv_std
1	150	0.8056	0.0046
2	200	0.8055	0.0026
3	250	0.8047	0.0020
4	300	0.8043	0.0038
10	600	0.8033	0.0046

```
Best n_estimators: 150
```

learning_rate

In [72]:
```
results_lr = []
best_lr = None
best_lr_score = -1

lr_list = [0.01, 0.03, 0.1, 0.2]

for lr in lr_list:

    xgb = XGBClassifier(
        random_state=42,
        n_estimators=best_trees,
        max_depth=best_depth,
        learning_rate=lr,
    )

    fold_scores = cross_val_score(
        xgb,
        x_train,
        y_train,
        cv=cv,
        scoring="accuracy",
        n_jobs=-1
    )

    mean_score = fold_scores.mean()

    results_lr.append({
        "learning_rate": lr,
        "cv_mean": mean_score,
        "cv_std": fold_scores.std()
    })

    if mean_score > best_lr_score:
        best_lr_score = mean_score
        best_lr = lr
```

In [73]:
```
results_lr_df = pd.DataFrame(results_lr).sort_values("cv_mean", ascending=False)
display(results_lr_df.round(4))

print("Best learning_rate:", best_lr)
```

	learning_rate	cv_mean	cv_std
2	0.10	0.8056	0.0046
3	0.20	0.8049	0.0044
1	0.03	0.7974	0.0076
0	0.01	0.7863	0.0032

```
Best learning_rate: 0.1
```

best model

In [74]:
```
best_params_xgb = {
    "max_depth": best_depth,
```

```
    "n_estimators": best_trees,
    "learning_rate": best_lr
}

print("Best XGB params:", best_params_xgb)

best_XGB = XGBClassifier(
    random_state=42,
    max_depth=best_depth,
    n_estimators=best_trees,
    learning_rate=best_lr,
)

best_XGB.fit(x_train, y_train)
```

```
Best XGB params: {'max_depth': 5, 'n_estimators': 150, 'learning_rate': 0.1}
```

Out[74]:

```
XGBClassifier
XGBClassifier(base_score=None, booster=None, callbacks=None,
              colsample_bylevel=None, colsample_bynode=None,
              colsample_bytree=None, device=None, early_stopping_roun
ds=None,
              enable_categorical=False, eval_metric=None, feature_typ
es=None,
              feature_weights=None, gamma=None, grow_policy=None,
              importance_type=None, interaction_constraints=None,
              learning_rate=0.1, max_bin=None, max_cat_threshold=Non
```

In [75]:

```
best_xgb_params = {
    "max_depth": 5,
    "n_estimators": 150,
    "learning_rate": 0.1
}

xgb_params_df = pd.DataFrame(
    list(best_xgb_params.items()),
    columns=["Parameter", "Value"]
)

display(xgb_params_df)
```

	Parameter	Value
0	max_depth	5.0
1	n_estimators	150.0
2	learning_rate	0.1

Results

In [76]:

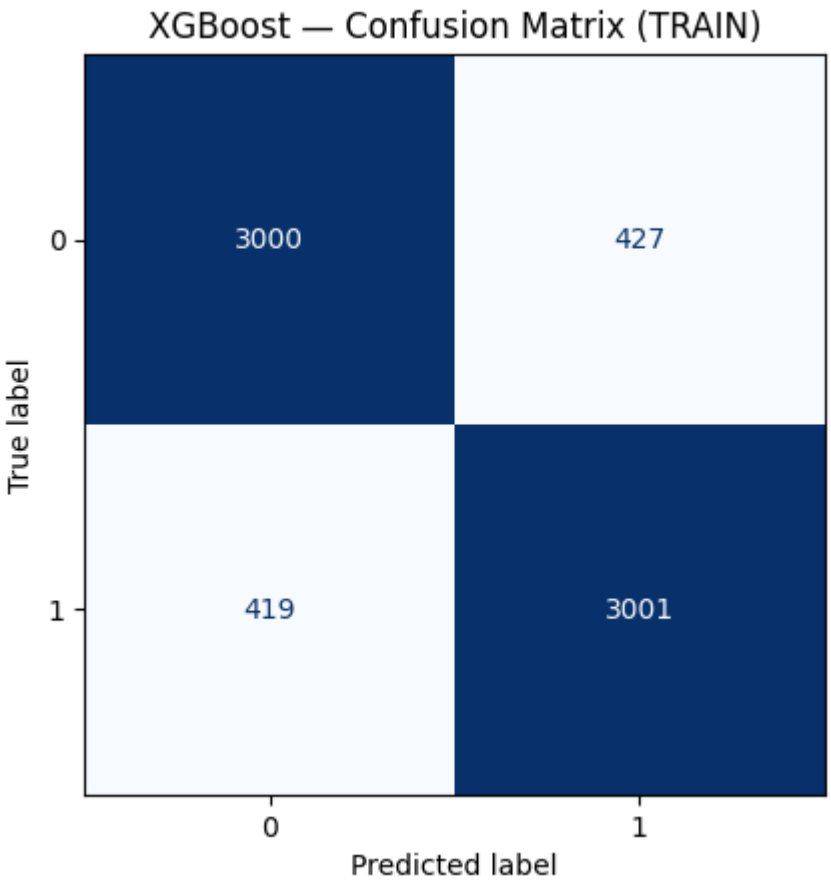
```
y_pred_train = best_XGB.predict(x_train)
train_acc = accuracy_score(y_train, y_pred_train)
print("="*40)
print("BEST XGB — TRAIN SET PERFORMANCE")
```

```
print("="*40)
print(f"Accuracy : {train_acc*100:.2f}%")
print("="*40)
```

```
========================================
BEST XGB — TRAIN SET PERFORMANCE
========================================
Accuracy : 87.64%
========================================
```

In [77]:
```
plt.figure(figsize=(5,5))
ConfusionMatrixDisplay.from_predictions(
    y_train,
    y_pred_train,
    cmap="Blues",
    colorbar=False
)

plt.title("XGBoost — Confusion Matrix (TRAIN)")
plt.show()
```

```
<Figure size 500x500 with 0 Axes>
```



In [78]:
```
acc = accuracy_score(y_train, y_pred_train)
prec = precision_score(y_train, y_pred_train)
rec = recall_score(y_train, y_pred_train)
f1 = f1_score(y_train, y_pred_train)

metrics_df = pd.DataFrame([{
    "model": "XGBoost",
    "accuracy": acc,
    "precision": prec,
    "recall": rec,
    "f1_score": f1
```

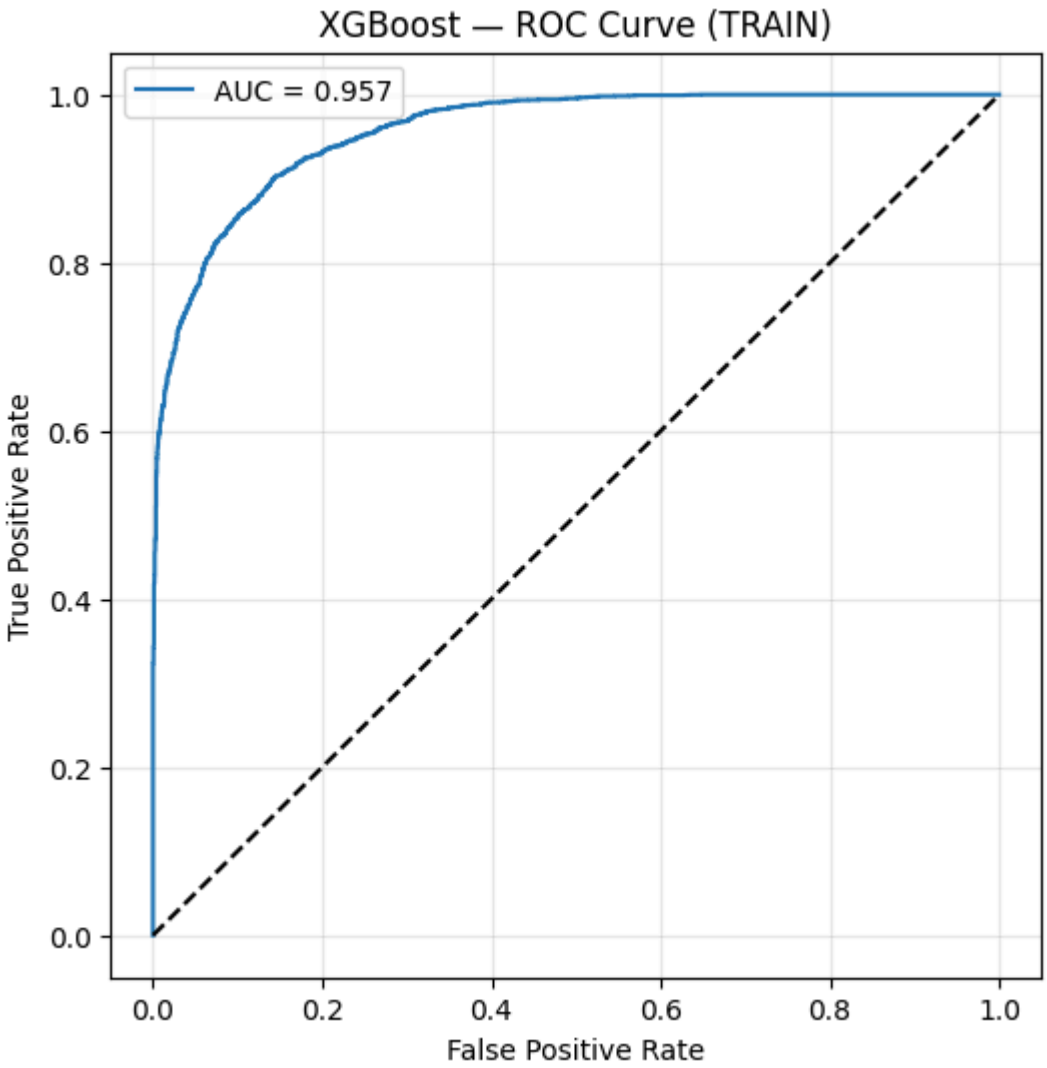
```
}])

display(metrics_df.round(4))

print("\nClassification Report (TRAIN):")
print(classification_report(y_train, y_pred_train))
```

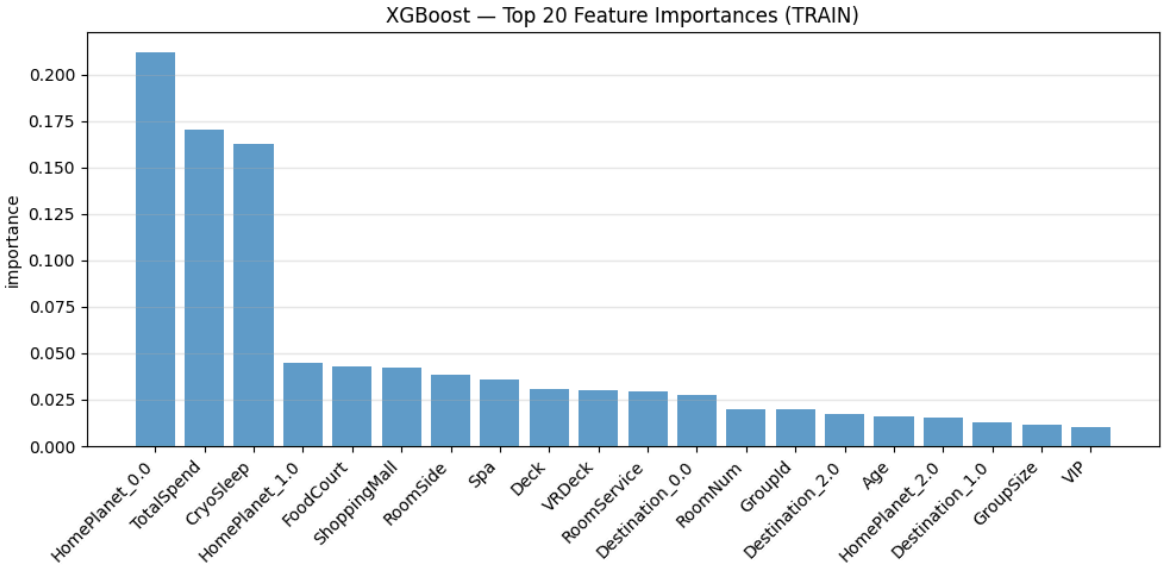
	model	accuracy	precision	recall	f1_score
0	XGBoost	0.8764	0.8754	0.8775	0.8765

```
Classification Report (TRAIN):
              precision    recall  f1-score   support

           0       0.88      0.88      0.88      3427
           1       0.88      0.88      0.88      3420

    accuracy                           0.88      6847
   macro avg       0.88      0.88      0.88      6847
weighted avg       0.88      0.88      0.88      6847
```

In [79]:

```
y_prob_train = best_XGB.predict_proba(x_train)[:, 1]

fpr, tpr, _ = roc_curve(y_train, y_prob_train)
roc_auc = auc(fpr, tpr)

plt.figure(figsize=(6,6))
plt.plot(fpr, tpr, label=f"AUC = {roc_auc:.3f}")
plt.plot([0,1], [0,1], "k--")

plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("XGBoost – ROC Curve (TRAIN)")
plt.legend()
plt.grid(alpha=0.3)
plt.show()
```



In [80]:
```
feature_names = x_train.columns.tolist()
importances = best_XGB.feature_importances_

fi_df = pd.DataFrame({
    "feature": feature_names,
    "importance": importances
}).sort_values("importance", ascending=False)

display(fi_df.head(20))

x = np.arange(len(fi_df.head(20)))
plt.figure(figsize=(10,5))
plt.bar(x, fi_df.head(20)["importance"], alpha=0.7)
plt.xticks(x, fi_df.head(20)["feature"], rotation=45, ha="right")
plt.title("XGBoost — Top 20 Feature Importances (TRAIN)")
plt.ylabel("importance")
plt.grid(axis="y", alpha=0.3)
plt.tight_layout()
plt.show()
```

	feature	importance
15	HomePlanet_0.0	0.212128
8	TotalSpend	0.170108
0	CryoSleep	0.162630
16	HomePlanet_1.0	0.045105
4	FoodCourt	0.042841
5	ShoppingMall	0.042487
13	RoomSide	0.038339
6	Spa	0.035864
11	Deck	0.030750
7	VRDeck	0.029991
3	RoomService	0.029228
18	Destination_0.0	0.027465
12	RoomNum	0.020037
9	GroupId	0.019578
20	Destination_2.0	0.017378
1	Age	0.016014
17	HomePlanet_2.0	0.015218
19	Destination_1.0	0.013124
14	GroupSize	0.011252
2	VIP	0.010363